Supplementary material for
Business Themes in the Trademark Record

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Section, table, and figure numbers without an S prefix refer to the main text. Sections S1–S5 carry the corpus and measurement appendices; S6 the material demoted from the main text's results sections; S7 the full robustness treatment, of which the main text's Section S8 is the summary. The interactive per-class appendix, including a viewer of every class's registrations on the lead/atypicality plane, is at `https://aporia.institute/tm-vocabulary/online-appendix/`.

S1 Corpus, scoring, and burn-in

S1.1 Corpus and scoring

The USPTO Trademark Annual Applications backfile (product TRTYRAP) provides a complete record of US trademark filings from 1884 onward. The deepest diagnostic work uses the complete software/electronics class (Nice 009; 1.81M filings); the diffusion application uses four classes totalling 2.32 million granted filings 1990–2024; the gate analyses use all 45 classes. The firm-level panels link trademark owners to SEC EDGAR financial filers and to PatentsView patent assignees by normalized-name join.

The theme representation is a $T = 50$ LDA fit on a stratified sample of 448,437 filings across all 45 classes (vocabulary of 62,168 unigrams and bigrams); every filing 1990–2024 is transformed to a dense 50-theme distribution, and past-facing and future-facing surprise are computed against per-filing theme references spanning 1,826 days on each side of the filing date, scoring 1995–2019. A filing whose window holds fewer than 500 same-class filings on either side is left unscored, since a thin reference inflates the divergence mechanically; that floor, together with the corpus edges, is why roughly two-thirds of filings carry a score. A term-level variant is retained alongside, on the same anchoring: token distributions (within-class document frequency ≥ 50) against per-filing windows of the same 1,826 days, Dirichlet-smoothed ($\alpha = 1$) over the class vocabulary. Because a goods/services description runs to a few dozen words, the divergence is needed only at the terms a filing actually uses, so the reference is carried as a running count vector and read sparsely. The term variant is used where vocabulary identity is the object (phrase transit) and as the comparison scoring throughout. Both representations are also computed under an exponentially decayed reference with a two-year half-life, truncated at the same window; Appendix S8.11 reports what changes.

S1.2 Scores are interpretable from 1995

Burn-in is the span at each end of the corpus where the reference windows cannot be filled, so scores there reflect a thin reference rather than the text being scored.

The signal depends on having enough past and future filings to populate the references; the 5-year past window is incomplete before 1990 and the future window after 2020, and both zones are excluded (Figure S1). The burn-in cut is set empirically: thin past references inflate past-facing surprise, and profiling that bias across all 44 sufficient-volume classes gives a median absolute bias of ∼2.0 nats in 1990, 0.25 in 1991, 0.10 in 1992, and 0.04 from 1993 onward, below any interpreted effect size. The 1995–1997 deviations (∼0.15–0.17) are not burn-in, because mechanical contamination declines monotonically as the windows fill and cannot rebound. They reflect a genuine vocabulary shift in the corpus during the dot-com period. Per-class burn-in lengths estimated under a stability-band criterion track each class's noise floor rather than its reference density, so a single standard cut at 1995 is used, conservative by about two years.

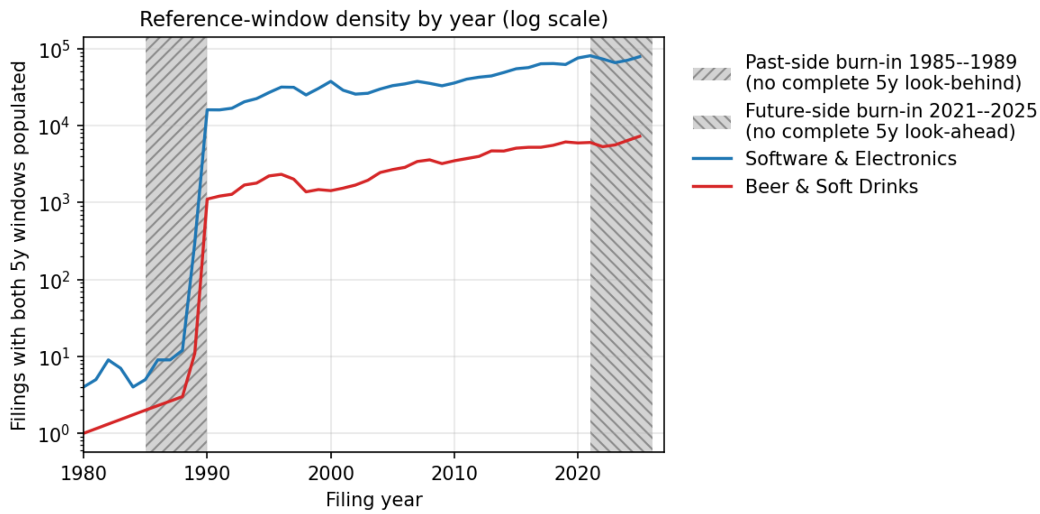

Figure S1: Reference-window density by year on log scale. Hatched regions are the past-side and future-side burn-in zones where the 5-year window is incomplete. The dotted line marks the 1995 analysis cut. The windows decide the score at the filing date; the gate outcome is read separately, at registration age six to seven.

S2 Examination rewards a middling specificity

Across 3.59 million debut filings (57.6% base rate), completion is an inverse-U in *atypicality*: it climbs from 46.6% at the least atypical end to about 58% in the middle and falls back to 53.4% at the most atypical (Figure S3, panel C). The same shape appears on the past-facing level alone, more mildly, running 54.0% at the bottom quintile to 58.4–59.1% in the middle and 57.7% at the top (Figure S2A).

The signed axis behaves differently. Completion on L falls to a trough of 51.9% just above the median and then rises steadily to 58.0% at the leading end. That trough is composition rather than timing: $|L|$ correlates with atypicality at +0.31, so filings near zero lead are disproportionately low-atypicality filings, which complete poorly for the reason above. Split by atypicality, the trough is absent in the upper three fifths and survives in the lowest. The signed axis still enters a linear specification strongly (Table 9), but monotonically rather than through an interior peak: what curvature exists at registration belongs to how far a filing's language sits from its category, not to which side of it the filing sat.

The obvious objection is that unfamiliar language is simply refused more often, so the measure recovers nothing beyond familiarity. Completion falls at *both* ends, so the most category-typical filings also complete less than the middle, which familiarity cannot explain; and the curvature is specific to the unsigned axis, the signed axis showing no interior peak at all.

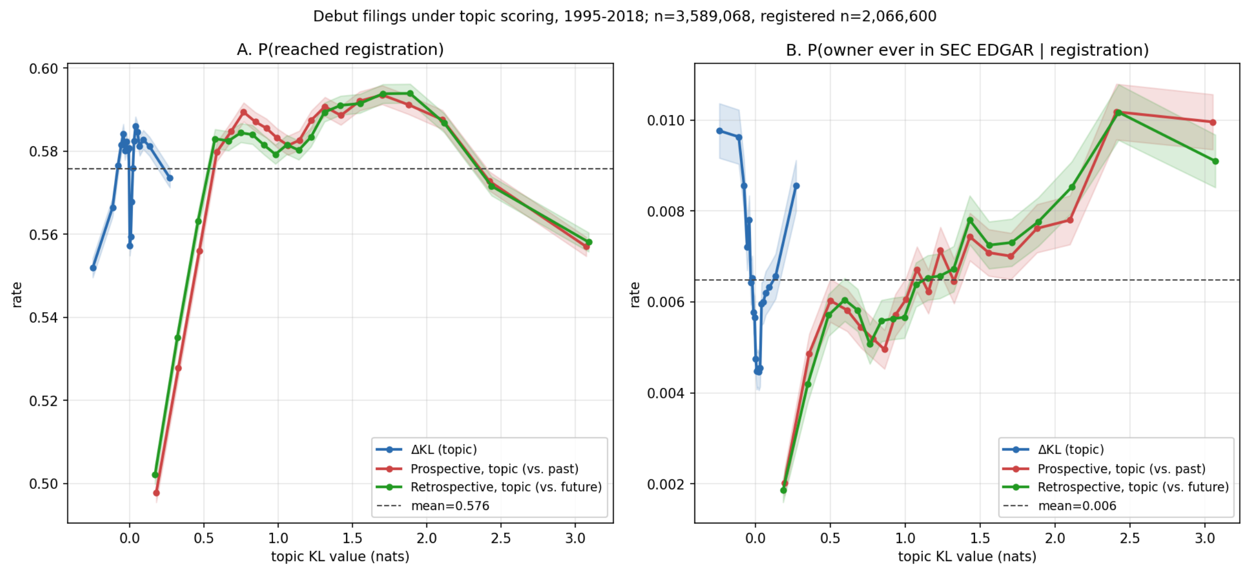

Figure S2: Outcomes for debut filings (owner's first-ever filing) across all 45 Nice classes under theme scoring, filing years 1995–2018, in quantile bins with 95% intervals, quantiles cut on the pooled distribution. $n = 3{,}589{,}068$; registered subset $n = 2{,}066{,}600$. *A:* P(reached registration). *B:* P(owner ever in SEC EDGAR, the Commission's public filing system | registration).

Refiling after abandonment is itself mildly related to vocabulary position, though not on the axis that matters here. Refiling rates are flat in atypicality (a middle-minus-tails difference of −0.06pp on a 3.6% base) but rise with lead, from 2.6% among the most lagging abandoned applications to 3.4% among the most leading.

S3 Internet-bearing filings by class

S4 How refiled descriptions are rewritten

Figure S5 shows the goods/services description before and after a rewrite, for the 31,084 refilings in Section 4.3 whose text changed. A pair is an abandoned application and the same owner's later registration of the identical mark in the same Nice class within six years; the earliest registered refiling is kept. Counsel of record is read from the prosecution history of each filing separately, so a pair can change representation between attempts. Atypicality is on the production scoring; length is the word count of the normalized description.

The means understate the convergence, because it runs in both directions. By fifth of the original's atypicality, the change is +0.16, +0.14, +0.06, −0.06 and −0.39 nats from the most conventional fifth to the most atypical, and within a point of that profile in each of the four representation groups. Identical-text refilings change by +0.002 to +0.008 across the same fifths. On lead, the changed-text profile runs +0.046 to −0.072 and the identical-text profile +0.014 to −0.054; the difference is the rewriting, the rest is the reference windows moving with the later filing date.

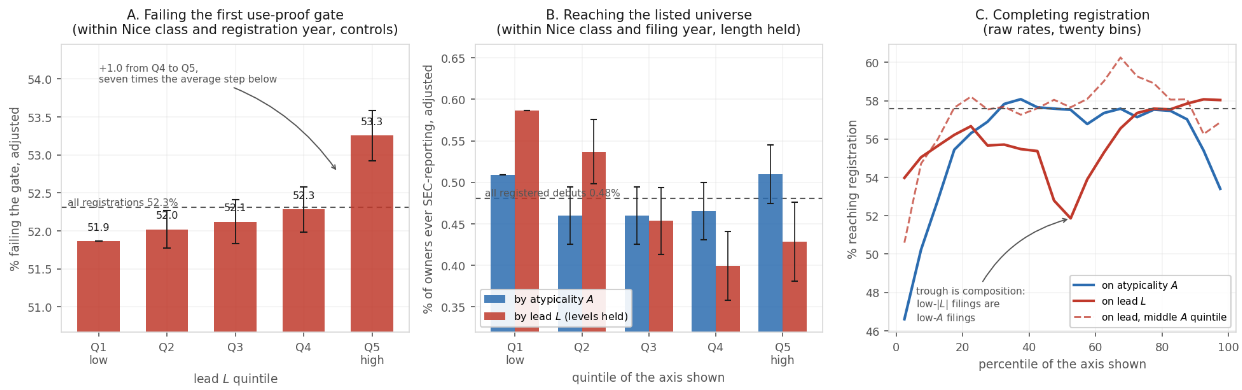

Figure S3: Quintile profiles for the three outcomes. *A:* the share of registrations failing the five-year proof of continued use, by lead quintile, adjusted within Nice class and registration year with the drafting and filer controls of Table 3; the dashed line is the rate for all registrations. The profile rises gently and monotonically through the fourth quintile and then steps up: about seven tenths of the top-to-bottom gap is the single step from the fourth quintile to the fifth, seven times the average step below it. *B:* the share of first-time filers whose owner ever reaches SEC reporting, by quintile of atypicality and of lead, adjusted within Nice class and filing year with description length held. *C:* raw registration rates for debut filings in twenty bins, cut on the pooled distribution — an inverse-U on atypicality, and on lead a trough just above the median that is compositional, absent once atypicality is held in its middle fifth (dashed). Horizontal line at the pooled rate.

S5 Term scoring measures length; theme scoring does not

Figure S6 puts the two representations side by side over the same axes, with each hexagon coloured by the mean length of the filings in it. Under term scoring the colour runs from dark near the origin — a median of about 200 terms — to near-white at the far end, where the median is eight. That gradient is the measurement problem: a filing scores as atypical largely because it is short. Under theme scoring the same surface is a narrow band of roughly uniform tone.

Two filings are marked in both columns. Under the production scoring they behave differently from one another, which is the point. Amazon.com (1997) is at the 98th percentile of lead on terms and the 94th on themes, but only the 40th and 47th percentiles of atypicality, so both agree that it was early and neither mistakes it for strange. Apple's iPod (2001) divides them: the 31st percentile of lead on terms against the 62nd on themes. The surfaces plotted here are on the class-year build the figure was generated from, where the length gradient is starkest; the percentiles just quoted are on the per-filing build used for every estimate.

Theme scoring is independent of document length, which term scoring is not; term scoring resolves novel word *combinations*, which theme scoring does not. Hence theme scoring for every estimate, since the length confound contaminates firm-level comparisons, and term scoring for every worked example, since the theme basis has no coordinate for the language the examples turn on.

S6 How three filings encode

Table S2 traces three filings through term scoring and theme scoring at $T \in \{50, 200, 500\}$ — three kinds of novelty, and how each representation handles them.

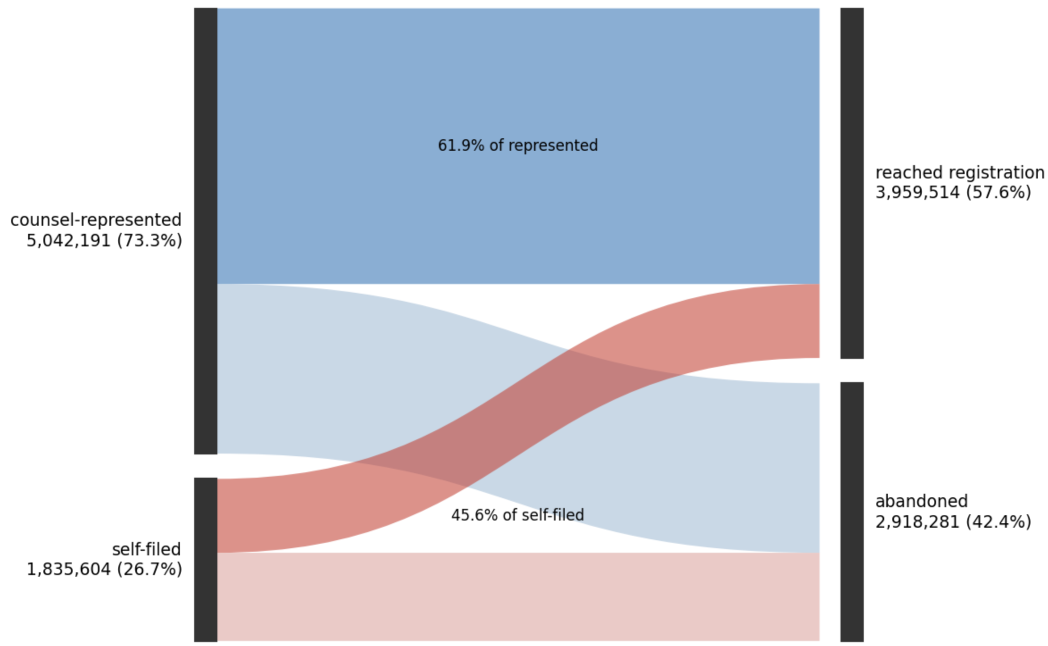

Figure S4: The registration step as a flow, unique serials filed 1995–2018 ($n = 6{,}877{,}795$), split by counsel of record on the filing. Represented applications reach registration at 61.9%, self-filed at 45.6%.

Amazon.com (1997, advertising/retail) is *combinatorial* novelty: every word is ordinary, but the bigrams assemble an invention ("computerized line," "line search," "search ordering"; class document frequencies 379, 134, 355). The filing's text reads "on line" as two words, the standard spelling in 1997, before *online* lexicalized into a single token; the analyzer drops the stopword "on" and the bigram "line search" survives stopword removal. The novelty was necessarily combinatorial because the word for it did not yet exist. Term scoring sees these bigrams; under theme scoring at every resolution all of them are out of vocabulary and the filing reads as a media retailer that uses computers.

Kombucha (1997, beer/soft drinks) is *lexical* novelty: the new thing arrives as new words (*kombucha* df 567, *saccharose* out of vocabulary even at the class level). The word *kombucha* is in the theme vocabulary but, with no kombucha-like theme, the filing loads on a generic beverages theme at every T (Table S2).

Ethereum (2018, software) is *wave* novelty: emergent vocabulary already diffusing (*blockchains*, *distributed computing*). It loads on generic software and services themes at every resolution, including $T = 200$, which does contain a cryptocurrency/blockchain theme; there, the eleven occurrences of *software* dominate the theme assignment.

The pattern across the row is the measurement point in miniature: theme scoring cannot represent combinatorial novelty (Amazon), dilutes lexical novelty (Kombucha), and majority-votes away wave novelty (Ethereum) even when a fitting theme exists. That the mark-level outcome results are nearly identical under both scorings despite this blindness is what licenses theme scoring as a robustness instrument; that the firm-level results diverge is what exposes them as artifacts of the lexical specificity term scoring additionally captures (Section S8.12).

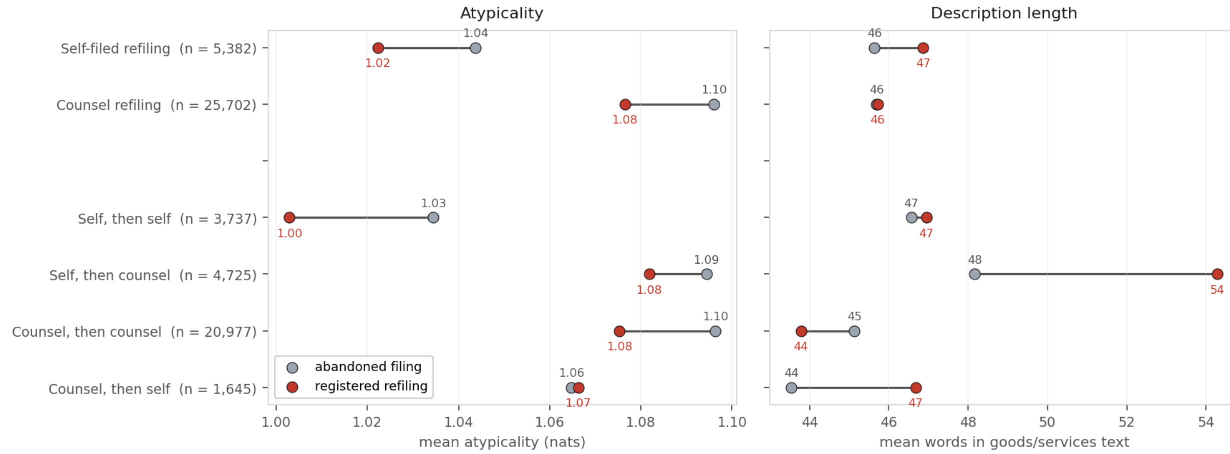

Figure S5: Rewritten refilings: mean atypicality and mean description length of the abandoned filing (grey) and of the registered refiling (red), by representation on the refiling and by the change in representation between the two attempts. Mean atypicality falls by 0.01 to 0.03 nats in every group except counsel-then-self, where it is unchanged. Length rises by six words where counsel is hired for the second attempt and falls by one where counsel rewrites counsel.

S7 Material demoted from the results sections

S7.1 Registration: refiling propensity

Refiling after abandonment is itself mildly related to vocabulary position, though not on the axis registration responds to: refiling rates are flat in atypicality (a middle-minus-tails difference of −0.06pp on a 3.6% base) and rise with lead, from 2.6% among the most lagging abandoned applications to 3.4% among the most leading.

S7.2 The five-year proof: timing, ties, and per-class extremes

The five-year proof is close to a fixed appointment rather than a hazard spread over years: of the 1.35 million cancellations recorded for fully observed cohorts, 17 post before registration age 6.5, 96.1% post between 6.5 and 7.0, and the rest after seven. Since the cancellation record ends on 2 April 2026, a cohort is observable only once it has reached seven years, which makes 2019 the last one with usable data at this proof and leaves 2020 and later unobserved. On the 3.15 million registrations whose ninth anniversary is inside the record — for which failure is settled and no window is needed — the contrast is +2.71pp within class and registration year, against +2.65pp on the paper's 4.0–8.5 window; conditional on failing, the probability of failing late carries a contrast of −0.01pp ($t = -0.2$). Response speed does not predict the outcome: across the 1.22 million scored registrations with a usable office-action/response pair, passage varies between 46.1% and 48.1% across response-speed quintiles, from a median of seven days to 183, and the quickest responders pass slightly *less*; the lead penalty is present in both halves (-3.48 ± 0.40 and -2.36 ± 0.40pp on passing, against −2.94 pooled). The restriction drops 61% of the sample, disproportionately applications that never drew a refusal.

Tied scores: within the class×registration-year cell the quintiles are cut in, 17.1% of scored registrations share a lead value with at least one other filing, and the largest tie group runs to 53. Filings that share a class, a filing date and a goods/services description have identical distributions scored against identical references. Cut points are fixed by sorting on the score and then on the serial number; sorting ties the opposite way moves the top-versus-bottom contrast by 0.001pp, and five pseudorandom orderings move it by at most 0.002pp, against a contrast of 2.287pp.

At the year-ten renewal, fitted over 872,079 five-year survivors with a resolved status, a straight line through renewal against lead explains 1% of the variation across forty bins and a V explains 89% (ΔAIC = 83); renewal peaks at 64.2% near $z = -0.4$ and falls to 45.5% at the most lagging bin and 50.6% at the most leading. In the software class, where the five-year lift on the 2010–2013 cohorts reaches +12.1pp, the conditional lift at the renewal falls to +7.6pp.

Per class, the largest raw penalties sit in tobacco (+26pp, the vaping era), hand tools (+10pp), processed foods (+10pp) and medical apparatus (+9pp), with negative lifts in beer and soft drinks, transport and construction; these are the extremes of a 44-class screen, sitting away from zero partly by selection.

S7.3 Waves: the theme tour, internet convergence, and surges

In 1995–1999 the computer, software, information and data services theme failed at 64.9% against 47.5–59.0% in the other large themes. In 2000–2004 it held 12% of the leading fifth; the leading language had moved to video, music and games on electronic media (31% of the leading fifth against 12% of filings), and inside several themes the lagging fifth failed more than the leading one: −3.6pp in electronic media, −5.5 in retail store services, −6.9 in apparatus and control systems. From 2008 the leading fifth over-represented downloadable and online software 15% against 4%; the theme's within-theme contrast cut inside the 2008–2014 era alone is +3.3pp, and the computer-services theme's own contrast is positive in every era (+3.6 to +5.9pp) and never reverses.

The internet gap in event time (Figure S7): dating the internet's arrival in each class as the first filing year the pattern reaches 2% of the class's filings (1994–1995 in the technology and transport classes, 1999–2000 in apparel and chemicals, 2008 in machinery), the gap between internet-bearing registrations and the rest starts near +7 points, declines to about zero nine to eleven years after arrival, and settles between +1 and +4; across the eight classes observable in both phases the mean gap is +4.4 points in years zero to four and +1.6 from year ten, shrinking in six of the eight. Calendar time disguises this: transport (arrival 1995) peaks at +16 points in its 2000 cohort; machinery (arrival 2008) peaks at +17 in its 2012–2015 cohorts. Clothing is the standing exception: its internet-bearing registrations outlast the class in every cohort after 2004. The per-class rates are tabulated in the internet appendix section above.

Surge episodes (Figure S8): measured against the whole corpus at a 1% share, five themes ever surge and one clears a 1,000-registration floor — online software, 1999, whose 6,339 in-window registrations failed at 63.7%. Within-class episodes differ in sign among themselves: the education-services surge in advertising and retail after 2002 failed 13.7 points *less* than its class; the charitable-fundraising surge in finance after 2011 failed 4.0 points more; of the episodes large enough to carry their own lead contrast, one is positive (+5.8pp, $t = 2.2$) and the rest indistinguishable from zero. On the 61 episodes with at least 300 joining registrations over a six-year entry window (64,105 of 86,447 joining registrations): joiners in the surge year fail −0.8 points relative to their wave's mean and joiners five years in +0.0, with no gradient between (SEs ~ 0.5); excess failure is +2.7, +3.9 and −0.1 points across wave-size terciles, correlating with size at −0.22; and waves whose theme share is still at or above its surge level five years on carry +4.3 points of excess against +1.0 for receding waves.

S7.4 Success: the unfunded listings, the value tiers, and the curated vocabularies

Of the 1,669 debut owners with an IPO marker, 759 have no post-debut Form D round in the record; their debut lead averages -0.22 standard deviations (SE 0.04) against -0.07 (SE 0.04) for the 910 funded, 29% sit in the most lagging fifth against 22%, and both groups are mildly atypical (+0.13 and +0.15). Rounds before 2009, foreign placements, and registered offerings post no Form D.

Value tiers: ten linked owners hold 18.5% of $19.2 trillion of linked peak revenue, fifty hold 41.5%, a hundred 53.6%, five hundred 82.9%, a thousand 92.6%. The ten largest sit 0.35 standard deviations below average atypicality and within 0.01 of average lead; the fifty largest 0.08 below and 0.13 above; the five hundred largest within about 0.1 on both; revenue-weighting the whole linked set gives -0.02 and 0.00.

Curated vocabularies: internet wording appears in 2.94 million class-filing rows, AI in 167 thousand, blockchain in 114 thousand; registration rates are 52.1%, 34.5% and 31.2% against 57.6% for all debut filings (the last two cohorts are young). On the 2009–2018 ladder under the rebuilt Form D match, AI debuts are funded at 7.8% and blockchain at 8.0% against 2.8% for internet and 1.6% overall, while at the IPO rung the gap disappears: 0.16% of AI debuts, 0.22% of blockchain, and 0.21% of internet debuts against 0.17% overall (1,896 AI debuts, 1,366 blockchain).

Patent sector split: on the coarse complements/substitutes sorting (eleven complement classes assigned from class definitions; sectoral build at 200 themes, firms with at least three panel years, 50 firms and 300 firm-years per cell), complement classes run from sixth to twenty-fourth of 27 by coefficient and disagree in sign; the only significant cells are paper and housewares, both negative, neither a complement class; on the wider 34-class retention the significant positives are transport and cosmetics and the significant negatives include medical devices and machinery.

S7.5 Pair lead corpus-wide

Extending the software-class pair-lead construction to every class with sufficient volume (43 of 45; the calendar-bucket and per-filing anchorings agree at $r = 0.983$ on identical filings and give the same class-009 contrast): pooled within class and registration year, the top-versus-bottom pair-lead contrast in failure at the five-year proof is +0.03pp raw ($t = 0.3$) and +0.48pp net of the filings' own lead and atypicality (owner-clustered $t = 4.2$, 1.30 million owner clusters). The effect is genuinely heterogeneous: significantly positive in 13 classes (largest in tobacco, spirits, and jewelry), significantly negative in 3 (largest in haberdashery and advertising and retail, -2.1pp), and era-unstable within both flagship classes. Table S3 carries the per-class estimates.

S7.6 The measure: Amazon's theme shares and the length null

Of the six themes Amazon's 1997 filing drew on, the share of class-035 filings drawing on each in the five years before and after: retail store services 42.6 → 50.3%, computer and information services 58.9 → 59.9, media and entertainment goods 9.9 → 11.3, downloadable and online software 2.3 → 4.1, printed matter 8.3 → 7.8, marketing of consumer goods 10.5 → 9.3 (a theme counts as drawn on at a tenth of the filing's weight or more). Synthetic filings whose words are drawn at random from the reference itself, maximally ordinary by construction, score 5.72 nats at three tokens and 1.65 at four hundred, tracking log(distinct terms) to within a fiftieth of a nat; the average real filing scores marginally below that null at its own length. The theme-pair windows behind the pair-lead measure are calendar-year buckets scored only when they hold at least 2,000 filings; word-level recombination counts adjacent pairs appearing fewer than five times in the class's prior five years while both words appear at least fifty times, with the share taken over such eligible pairs; the raw top-versus-bottom recombination contrast is 6.1 points before the lead and atypicality controls reduce it to 2.8.

S8 Robustness: what changes under other choices

Each result above is stated under one formulation chosen for legibility: one theme resolution, flat weighting inside the reference windows, symmetric five-year windows, estimates conditional on grant, and the 2002–2018 registration cohorts. This section reports what each of those choices costs, and in each case states the more complicated finding beside the simpler one the body uses.

S8.1 Representation and weighting

A scoring is fixed by two choices. The *representation* is what the language is turned into: a distribution over 50 fitted themes, or a distribution over the class vocabulary's terms. The *weighting* is how days inside the reference window count: every day equal, or decaying by half every two years. Every reference is per-filing, anchored on the filing's own date (Section 3). Four builds exist:

Representation	Weighting	sd(lead)	sd(atypicality)
Themes	flat	0.102	0.698
Themes	half-life 2y	0.082	0.698
Terms	flat	0.155	1.101
Terms	half-life 2y	0.130	1.102

All four are compared on the 6.01 million filings every one of them scored. The first row is the production scoring.

Changing one choice at a time settles the ordering. Correlations of lead between builds differing in exactly one respect:

Choice varied	r (lead)	r (atypicality)
Weighting, flat vs half-life 2y (themes)	0.982	1.000
Weighting, flat vs half-life 2y (terms)	0.970	1.000
Resolution, 50 vs 200 themes[1]	0.636	0.828
Representation, themes vs terms	0.460	0.396

The weighting is close to irrelevant: whether recent language counts for more than distant language within the window barely moves the score, and it does not move atypicality at all to three decimals, which is what the 45-degree rotation of Section 3 implies — the long axis is a property of the filing and the weighting acts almost entirely on the small residual. Resolution and representation matter most: quadrupling the number of themes leaves the two builds agreeing on 40% of the variance in lead, and themes against terms on less than a quarter. A robustness check that varies only the weighting is therefore close to a restatement; one that varies the representation or the resolution is a real test, which is why the term-scored replication in Appendix S8.11 and the sweep below are the ones the claims lean on.

[1]The resolution row is computed on the five-year-proof sample rather than the 6.01 million-filing common sample of the other rows.

S8.2 Theme resolution

Fifty themes over a 62,168-term vocabulary is roughly one theme per industry; whether the findings depend on that coarseness is testable by refitting at higher T. Raising the theme count does not add coverage — the vocabulary is fixed, and a larger T re-partitions the same words — but it changes what counts as sitting far from an industry, and could change any sign. Re-fitting the model at four resolutions on the same stratified sample and the same analyzer, and re-scoring the software class under per-filing windows:

Themes	Proof contrast, lead	t	Proof contrast, atypicality	r with $T = 50$
50	+3.23pp	13.8	−14.01pp	—
100	+2.71pp	11.6	−13.69pp	0.690
200	+5.20pp	22.2	−13.45pp	0.640
500	+5.69pp	24.3	−14.49pp	0.641

Three readings, in descending order of how much weight they carry. The sign and the significance are untouched across a tenfold change in resolution, and so is atypicality, which stays within a point of −14pp throughout. The magnitude is not: it does not move monotonically in T, so no single resolution can be quoted as the software class's effect, and the figure this paper reports at $T = 50$ sits at the low end of the range rather than the high one. A fifth fit at $T = 1000$ exists and is excluded: its first run shipped undertrained models through an indentation bug, and the refit is not persisted. And agreement with $T = 50$ settles at 0.64 without decaying as T grows, which says the resolutions share a fixed core and differ in the remainder rather than drifting apart as the partition refines.

The corpus-wide contrast is steadier than the single class: at $T = 200$, scored for all 45 classes, it is +2.23pp against +2.29pp at $T = 50$. Only those two resolutions exist corpus-wide, so that is a two-point comparison, but it is the one the paper's estimates run on, and pooling evidently averages away much of the per-class resolution sensitivity.

S8.3 Fitting noise at one resolution

The sweep leaves one question open. Scores at different resolutions agree at $r = 0.64$ to 0.69 on lead, and neighbouring resolutions agree no better than distant ones. That has two readings. The number of themes may genuinely change what is being measured. Or a theme-model fit may be unstable, with two runs settling in different local optima, in which case the sweep measures fitting noise and the resolution has little to do with it.

Holding T fixed and moving only the seed separates the two. A second fit at $T = 200$ on the same sample, the same vocabulary and the same number of passes, differing from the first in nothing but its random seed, agrees with it at $r = 0.79$ on lead and $r = 0.87$ on atypicality across 1.11 million scored software-class filings. The cross-resolution figures were 0.64 and 0.83. Most of what the sweep showed as resolution sensitivity is therefore present between two fits that differ only in the seed: for atypicality nearly all of it, for lead well over half. The resolutions are not measuring different things; each fit is a noisy draw on the same thing.

The estimate moves less than the scores do. On the identical sample, the five-year-proof lead contrast is +5.20pp under the first seed and +4.88pp under the second (SE 0.23 each), and the atypicality contrast −13.45 and −14.38pp. Two consequences follow. A single fit's lead score carries measurement error of roughly a fifth of its variance, which attenuates every coefficient on lead toward zero; correcting for it would enlarge the estimates, not shrink them. And the spread of magnitudes across the resolution table above, which does not move monotonically in T, should be read as a band about a third of a point wide from fitting noise alone rather than as a property of any particular T.

The substantive claim does not depend on any of this. The five-year-proof lead contrast on the 2002–2018 registrations each build scores runs +2.29, +1.95, +3.56 and +2.92pp across the four builds in the order tabulated above, and the atypicality contrast −4.39, −4.43, −3.63 and −3.73pp. Every sign holds; term scoring returns the larger lead contrast, so the headline estimate is the conservative representation.

S8.4 Decayed reference windows

Inside the reference windows every day counts the same. The alternative weights recent language up, with a half-life of two years inside the same 1,826-day truncation, so that a term the class adopted last month is more familiar than one it adopted four years ago. Rescoring the corpus that way changes the scores little and the estimate somewhat. Flat and decayed lead correlate at 0.980 on the 3.10 million gate registrations and atypicality at 1.000 to three decimals. On the identical registrations, with quintiles cut within class and registration year, the top-minus-bottom gate contrast on lead is +2.29pp under the flat window and +1.95pp under the decayed one (SE 0.09 each); on atypicality it is −4.39 and −4.43pp. The decayed window therefore returns a lead penalty about a seventh smaller, with the same sign, the same precision, and the same profile. The flat window is kept in the body because it is the simpler statement; a reader who prefers recent language to count for more should take +1.95pp as the estimate.

S8.5 Window width

The window is a parameter of the measure, not a property of it, and the five years used throughout was inherited from the class-year construction rather than chosen under the current one. Re-scoring the software class at several widths under per-filing references, and judging each by the five-year-proof quintile contrast it produces, gives +2.02pp at two years, +2.20 at three, +3.23 at five and +3.37 at seven, each on an identical sample of 452,911 registrations and each with a standard error of 0.23pp. Signal rises steeply from two to five years and is flat between five and seven: the +1.03pp gain from three to five is well outside the sampling error, the +0.14pp from five to seven is not. A ten-year window returns +2.89pp, but that figure is not comparable to the others — requiring both windows to lie inside the corpus costs 22% of the scored filings and 13% of the gate sample, dropping the earliest and latest registration years, which are also the years whose contrasts differ most. The apparent decline at ten cannot be separated from that change in composition and is not evidence about window width. On the range where the sample is fixed, seven years is nominally the peak and five is within noise of it while retaining more of the corpus, so five is defensible rather than optimal. The sweep is on one class and one outcome.

S8.6 Mixing the two windows

"Leading" is relative to a reference, and the two references need not be the same width. A short past and a long future make lead a measure of foresight — whether the filing resembles where the class was heading over the next seven years more than what it wrote in the last three. A long past and a short future make it a measure of rupture — whether the filing has broken with what the class had established over seven years, judged against only the next three. Scoring every class on the production representation and per-filing windows at $W \in \{3, 5, 7\}$ years on each side, and evaluating all five variants on identical samples — the gate columns on the 3.10 million registrations

2002–2018, quintiles within class×registration-year; the registration columns on the scored filings 1995–2018, quintiles within class×filing-year:

Variant	Past, future (years)	First-gate failure Q5−Q1 (pp)	First-gate failure classes > 0	Registration Q5−Q1 (pp)	Registration Q3 − tails
Foresight	3, 7	+1.65 (0.09)	31 of 45	+1.09	+0.50
Symmetric	3, 3	+1.69 (0.09)	—	+0.79	+0.13
Symmetric	5, 5	+2.29 (0.09)	33 of 45	+0.71	+0.28
Symmetric	7, 7	+2.58 (0.09)	—	+0.48	+0.65
Past-rupture	7, 3	+3.00 (0.09)	37 of 45	+0.01	+0.33

The symmetric five-year row is the paper's FE-only estimate in Table 3, reproduced here to the decimal, which fixes the scale. The penalty rises monotonically along the mix, from +1.65pp under foresight to +3.00pp under rupture, and rupture is the more pervasive variant, positive in 37 of 44 classes against 31. In software the two are +1.82 and +5.07pp. So the use requirement punishes distance from the class's established language more than it punishes resemblance to its coming language; but foresight carries a penalty of its own, more than half the symmetric one, and the separation is a matter of degree rather than kind.

Registration is a different matter. On the signed axis the top-to-bottom contrast is at most a point on a 57% base and the interior-minus-tails depth at most two thirds of a point, under every mix. The examiner's response to lead, on this scoring, is close to nil.

S8.7 Without conditioning on grant

Figure S9 repeats the lifecycle outcome analysis without the registration conditioning, on filings 2013–2015 pooled across all classes (a window whose registrations land by ∼2017 and face at least one Section 8 gate by the 2025 snapshot; the earliest registrations in the window also face their year-ten renewal, so panel B mixes one and two gates). Panel A is the unconditional composite, the probability that a filing both registers and passes the gate, measured over all filings including the never-registered. It is nearly flat in L because two opposed gradients cancel: registration rises in lead (61.3% to 63.2%) while gate survival falls (44.2% to 40.8%); the caption carries the quintile figures. Conditioning on grant is what separates them. Panel B shows the conditional gate outcome on the same filings, and it is U-shaped rather than inverse-U (depth −2.11pp), with a clearly penalised leading tail consistent with the cleaner single-gate cohort of Figure S10.

S8.8 Toggled filters, windows and parameters

Each new analysis in Sections 4–7 was rerun under its natural parameter toggles, on the identical data. Nothing reverses a headline claim; two auxiliaries move.

Variant	Baseline → variant	Reading
Gate window 3.5–9.0 / 4.5–8.0	lead +2.29 → +2.29 / +2.28pp	window immaterial
Registration cohorts 2002–09	+2.29 → +0.98pp	the era swing
Registration cohorts 2010–18	+2.29 → +3.11pp	penalty concentrated late
Atypicality, both cohort halves	−4.39 → −4.55 / −4.28pp	stable
Internet pattern, narrow core	goods web gap 0.0 → +2.0pp	divergence holds
Convergence arrival 1% / 4%	early→late +4.4 → +1.6	+5.9 → +1.8 / +3.8 → +2.3
Surge doubling 1.5× / 3×	in-surge net +1.3pp	+1.2 / +0.9pp
Wave entry window 4y / 8y	order flat	flat under both
Recombination floor 2 / 10	net −2.8pp	−1.2 / −3.6pp
Pair-lead presence 0.05	net +2.3pp	+3.1pp
Combination measures, class 035	009 values	recomb −3.8; pair-lead −1.4
Ladder, counsel / self debuts	reporting lead −0.19pp	−0.26 / −0.03pp
Ladder, 2009–13 / 2014–18	reporting atyp. +0.11pp	0.00 / +0.19pp
Excluding duplicate-text filings	lead +2.29; atyp. −4.39pp	+3.05; −5.68pp
Excluding Manual-scale texts only	same	+2.64; −5.05pp

Three of these change what a reader should carry. The lead penalty at the gate belongs to the 2010–2018 cohorts; the 2002–2009 half carries a third of it, which is Section 6.1's era swing seen from the other side. The ladder's gradients belong to counsel-represented debuts — among self-filers the reporting gradient is a null — and its atypicality gradient to the 2014–2018 half. And the pair-lead penalty, +2.3pp in the software class under every parameter tried there, is −2.1pp in advertising and retail on the corpus-wide construction, and pooled across 43 classes is zero raw and +0.5pp net of lead and atypicality, so it is reported as a bound for one class rather than a general fact. All artifacts are under `paper/v3/_eval/`.

S8.9 Changing the time period

The registration cohorts the gate estimates use begin in 2002 because that is the conventional cut for this corpus, and end in 2018 because the cancellation record runs to April 2026. Section 6.1 carries the series back to the first scoreable cohorts: the penalty was two to three times larger for marks filed in the late 1990s than for any cohort since, reversed for filings made 2000–2004, and returned from 2008, with the swing inside the four technology classes and, within them, in which themes each era's leading filings were drawn from. Within the 2002–2018 window the penalty is zero to negative for the 2002–2005 cohorts (−2.3pp for 2003) and runs +1.8 to +5.4pp for every cohort from 2008 on the wide-window raw series. The 2019 cohort, evaluable only on the narrow 6.5–7.0 observation band for registrations issued early enough in that year (78,408), is consistent in sign at +5.6pp. Cohorts from 2020 have no observed proof. The pooled estimate is therefore an average over a penalty that was not constant, and a reader who wants the penalty for a period should read Figure 5 rather than the pooled row.

S8.10 Lead is a small residual

Past-facing and future-facing KL are highly correlated however scored. On the complete class-009 corpus under the production scoring they correlate at $r = +0.988$ (n = 1,109,643). The lead L is therefore a small residual of two large, stable quantities: both levels have a standard deviation near 0.70 nats about a mean of 1.46, while L has a standard deviation of 0.109 — about a sixth of either level's spread. Only 1.2% of the two levels' combined variance survives into their difference. That is what makes L sensitive to anything that inflates the levels; under term scoring the dominant such thing is document length, and Section 3 quantifies the coupling under both representations.

The consequence for the lead is direct: the standard deviation of L rises 2.2–2.5-fold across deciles of A under term scoring and 1.5–1.8-fold under theme scoring, so a unit of lead means roughly the same thing everywhere only in the theme representation. Appendix S5 shows this as a length surface over the two axes. Two qualifications. Theme scoring returns a finite value for about 65% of filings, so the two columns are not drawn from the same set; re-estimating the term-scored correlations on exactly the theme-scored subset barely moves them, which puts the contrast down to the representation rather than the sample. And "flat" means no monotone length gradient rather than constant: class 035 under theme scoring is mildly U-shaped in A. The two scorings agree only moderately per filing (Pearson 0.46, Spearman 0.42 on the 6.01 million filings both score), which gives their agreement on the mark-level patterns evidential force and their disagreement on the firm-level patterns diagnostic force (Section S8.12). Across $W \in \{3, 5, 7\}$, 6.9–10.4% of filings flip sign. Per-filing noise attenuates at every aggregation used: firm-year means, token pools, theme prevalence trajectories.

Alignment with expert innovation judgment is at most suggestive. Firms on BCG/MIT "most innovative" lists sit +0.09 standard deviations above the panel mean under term scoring ($p = 0.22$), +0.14 under theme scoring ($p = 0.07$), and +0.08 at $T = 200$ ($p = 0.20$). The matched sample is 41 firms, which is too small to carry weight in either direction, and the paper draws no inference from it. Patent counts remain uncorrelated with L on the same panel (Pearson +0.010).

S8.11 Term scoring as a cross-check

The snapshot cross-check on the single one-gate-elapsed cohort (registrations 2016–2018, terminal status 710 versus 701–705/800) agrees with the event-dated estimates and is scoring-robust: −5.4pp survival across quintiles under theme scoring and −5.7pp under term scoring on the identical sample (Figure S10). Under term scoring, gate survival *rises* in the raw KL level (+6.5pp past-facing, +7.6pp future-facing): atypical text survives the use requirement better while leading text survives it worse — the same two-axis separation as Section 7.3, at the mark level.

S8.12 Term scoring at the firm level

Under term-level scoring, leading filings appear to belong to better firms. Among registered debut filings, the probability that the owner ever appears in SEC EDGAR rises in term-L from 0.29% at the bottom quintile to 0.42% at the top; on an SEC-matched firm-year panel, trailing 3-year mean term-L predicts gross margin at +2.2pp per standard deviation ($t = 2.7$ on the locally rebuilt panel through 2019; +3.2pp, $t = 5.3$, on the published panel through 2024), with past-facing surprise entering positively and future-facing negatively; and firm-year term-L predicts excess stock returns of +1.5 to +5.0pp per standard deviation at one-to-four-year horizons. A larger debut-only return figure (+28pp at 4 years) fails a sample-size diagnostic and is not claimed.

None of the signed versions survives distribution scoring. On identical samples, the debut listing gradient in signed theme-L is flat to declining, and the margin coefficient reverses: −1.3pp per standard deviation ($t = -1.8$) at $T = 50$, −2.2pp ($t = -3.1$) at $T = 200$, and −0.8pp ($t = -1.1$) at $T = 500$, against the term-level +2.3pp; the past-/future-facing decomposition flips sign and stays flipped. The term-level listing gradient separately dissolves under document-length controls: within length bands it is U-shaped rather than rising, with the monotone rise contributed by composition across lengths (listed firms' counsel write longer descriptions). The mark-level results are unaffected — they hold under both scorings at every resolution.

Vocabulary blindness does not rescue the signed signal. The theme model's vocabulary is sample-built, so rare and emergent terms are invisible to it; if the firm signal lived exactly there, theme scoring would erase a real signal rather than an artifact. Computing each filing's invisible-vocabulary share (63.8% of class-009 term *types* but only ∼9% of token *mass*): the share itself is a strong firm marker (P(EDGAR) 2.9% to 5.5% across its quintiles; gate survival +5.7pp), but *within* invisible-share strata the signed L gradient on the firm outcome is flat to negative, while the gate penalty holds in both strata (−6.9 and −8.3pp). The term-level firm signal is *unsigned specificity*: serious firms describe specific products with specific, therefore rare, words, and rare-term mass inflates term-L mechanically. Theme count is a partition dial, not a coverage dial — a crypto/blockchain theme exists at $T = 200$ but not at $T = 50$ or $T = 500$, because raising T re-partitions the fixed vocabulary along dominant axes — so resolution sweeps test granularity, not coverage, and the invisible-share split is the test that binds.

The registration filter does not explain the divergence. Dropping the registration conditioning leaves the term-level listing gradient essentially unchanged (0.26% to 0.38% across quintiles against 0.29% to 0.42% conditional), and an examiner channel touching 4.3% of failed applications cannot generate several-point quintile gaps.

Term-resolved text-novelty measures can therefore manufacture firm-performance correlations from drafting style and lexical specificity, and a distribution-based rescoring is a cheap, decisive diagnostic.

S8.13 Functional form

Four shapes were fitted to each profile by weighted least squares on 40 equal-count bins (Figure S11): linear, quadratic, an exponential $a + be^{cx}$, and a free-vertex symmetric form $a + be^{c|x-x_0|}$ meant to catch a U directly. Selection is by AIC.

Outcome	Axis	R^2 line	R^2 curve	ΔAIC	Shape selected
Registration	atypicality	0.38	0.92	77.1	curve, peak at +1.04
Registration	lead	0.08	0.17	1.9	(quadratic; weak)
Use-proof	atypicality	0.74	0.74	0.0	linear
Use-proof	lead	0.44	0.55	4.3	V, vertex −1.50
SEC reporting	atypicality	0.74	0.84	15.2	curve, peak at +2.17
SEC reporting	lead	0.13	0.59	26.3	V, vertex +0.30
listed only	lead	0.19	0.46	12.2	V, vertex +0.41
reporting, not listed	lead	0.07	0.67	37.2	V, vertex +0.24

The winning form is a free-vertex $a + b\,e^{c|x-x_0|}$. On the atypicality rows it is a genuine curve, with fitted exponents between 0.5 and 1.2 — a hump that peaks about one standard deviation above mean atypicality at registration and about two above it at SEC reporting. On every lead row the exponent collapses to the order of 10^{-4}, which turns $e^{c|x-x_0|}$ into $1 + c|x - x_0|$: what wins there is a *V*, linear in distance from a vertex. A V beating a parabola says the turn is sharper than a quadratic allows, not that anything is multiplying.

Vertices are in standard deviations of the axis across filings.

The two rows for SEC reporting are the outcome Section 7.3 and Table 8 actually run on; the listed and reporting-only splits below them show the same shape on each half, which is why the split does not change the reading.

AIC here is computed on 40 binned means, not on the millions of underlying filings, so it ranks shapes rather than testing them. The 77 at registration and the 12 to 36 at the public-reporting gates are decisive; the 1.9 at registration on lead and the 4.3 at the five-year proof are weak, and those two rows are better read as "no worse than linear" than as curvature.

The non-monotone profiles on signed lead decompose into parts of different symmetry (Figure S12). Each ventile's deviation of the listing rate from the base is split, by sequential counterfactual prediction, into the part atypicality composition alone predicts, the part exact-duplicate text adds, a linear lead tilt, and a residual. The symmetric U is carried about half by atypicality composition — the spread of L rises with A, so both tails of the lead axis hold atypical filings and the middle holds typical ones — and about half by a residual that is itself symmetric in $|L|$: the same conditional lead magnitude effect Table 8 estimates at +0.067pp per standard deviation. Duplicate-text composition adds almost nothing once atypicality is held, and the linear tilt is an order of magnitude smaller than either symmetric part (−0.07pp per nat). The fitted vertices above follow from this arithmetic: a symmetric profile plus a negative tilt has its minimum shifted right of zero, and a survival hump its maximum shifted left, which is where the fits put them (+0.24 to +0.41, and −0.4, standard deviations).

Excluding text-duplicate filings strengthens rather than weakens every headline gradient. Of the five-year-proof sample, 47% of registrations share their exact normalized description with at least one other filing; dropping them raises the lead contrast from +2.29 to +3.05pp and the atypicality contrast from −4.39 to −5.68pp, and dropping only the Manual-scale texts (clusters of 100 or more identical filings) gives +2.64 and −5.05pp. Identical texts carry identical scores and cluster at the typical end of the axis, so they dilute both gradients; neither is created by them.

S8.14 Listing against reporting

"Appears in SEC records" mixes two populations: companies whose shares trade, and entities that file with the Commission without ever listing — private issuers, funds, subsidiaries registering securities. Splitting the 19,889 matched owner strings on whether their CIK carries a ticker in the Commission's own company-ticker file gives 10,556 listed against 9,333 that file financial statements without one.

It does not change the finding. Among registered debut owners whose debut filing is scored, 0.310% later appear as listed companies and 0.285% as reporting non-listed ones, and both profiles are U-shaped in lead with vertices at +0.41 and +0.24 standard deviations. The listed profile is slightly flatter and noisier, which is what the smaller and more selected outcome predicts. Whatever the signed axis is picking up at this margin, it is not an artifact of counting non-operating registrants as commercial successes, and it is not specific to the act of listing either.[2]

S8.15 Filings are not independent draws

Some of them share an exact position with another filing, and outcomes cluster within owner — the within-owner correlation is 0.38 for gate survival pooled and is mechanically 1.0 for public listing, since listing is a property of the owner and a firm with hundreds of marks contributes hundreds of identical successes. The regression tables are unaffected, because the gate regressions cluster on normalized owner and the firm-level specifications carry one observation per owner. The raw quintile contrasts of Sections 5–6 — the era, theme, internet and surge tables — are not: their two-sample standard errors ignore the owner clustering, and their t-statistics overstate precision by roughly a factor of two, which the text of those sections now carries where the margins are close. Binning filings without the correction likewise overstates how much structure the corpus contains:

[2] Two limits on the split. The ticker file is a current snapshot, so a company that listed in 2003 and was acquired in 2011 appears in the non-listed tier; the tiers are therefore "currently listed" rather than "ever listed," which will misclassify older successes downward. And the base rates here are computed on the appendix's own panel, which conditions on the debut filing carrying a score and so is smaller than the body's sample.

on a positional grid, correcting cell errors for the clustering design effect — the factor by which within-owner correlation inflates the variance of a cell mean — cuts apparent excess dispersion by roughly a quarter on the gate outcome and by half on listing. Displays of the vocabulary plane are descriptive here for that reason, and the near-zero pooled correlation between A and L reported in Section 3 does not survive cell by cell within class, so per-cell contrasts should not be read as within-class effects.

S8.16 What remains unresolved

What drives the episodic pattern is not settled. Cohort and gate year move together, so post-crisis bet composition, app-era speculative filings, and calendar-time enforcement changes cannot be separated on this design. The representation is also fit on the pooled corpus; a vintage refit would close the remaining look-ahead channel.

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Table S1: Internet-bearing filings against the rest of their Nice class (Section 6.3). Registration is the share of class-records filed 1995–2018 that reached registration; failure is the share of unique registrations 2002–2018 cancelled for non-use at age 4.0–8.5 years. Classes with fewer than 500 internet-bearing registrations are omitted.

Class		Web share (%)	Registration (%) web	Registration (%) rest	Gate failure (%) web	Gate failure (%) rest
001	Chemicals	1.8	65.0	64.3	46.1	39.1
003	Cosmetics & Cleaning	2.8	58.0	54.2	52.1	54.4
004	Lubricants & Fuels	4.6	63.8	56.3	50.9	47.9
005	Pharmaceuticals	1.9	53.3	48.6	52.4	53.5
006	Metal Goods	3.5	68.1	67.0	44.5	39.1
007	Machinery	2.8	73.3	69.7	50.9	38.7
008	Hand Tools	3.6	65.8	62.5	42.4	43.6
009	Software & Electronics	18.7	53.3	57.0	56.5	51.4
010	Medical Apparatus	3.0	59.8	58.4	51.7	45.4
011	Lighting & Heating	3.0	67.4	64.8	51.4	45.9
012	Vehicles	3.4	69.0	60.9	49.6	45.1
014	Jewelry	7.4	63.2	55.8	54.7	55.7
016	Paper & Printed Goods	15.1	57.8	56.1	47.9	50.7
017	Rubber & Plastics	2.2	70.7	67.6	44.1	38.5
018	Leather Goods	8.0	61.4	57.4	50.5	55.4
019	Building Materials	2.0	71.4	64.0	47.5	43.9
020	Furniture	4.2	62.2	59.2	47.0	48.7
021	Household Utensils	5.1	60.1	59.0	44.5	50.5
024	Textiles	6.0	62.8	55.9	49.7	51.2
025	Clothing & Footwear	5.6	56.1	49.6	52.4	58.7
026	Lace & Embroidery	7.6	61.2	55.7	52.0	53.9
027	Carpets	5.6	66.8	56.8	43.7	46.9
028	Games & Sporting Goods	6.3	55.1	53.6	54.1	55.7
029	Meats & Processed Foods	2.5	59.7	55.7	53.3	49.6
030	Staple Foods	2.6	61.4	56.0	52.3	51.8
031	Agricultural Products	2.5	60.9	58.0	49.5	43.8
032	Beer & Soft Drinks	2.4	58.7	49.1	56.2	53.3
033	Alcoholic Beverages	1.2	61.3	52.7	56.9	42.3
034	Tobacco	2.9	53.8	49.1	55.0	54.2
035	Advertising & Retail	40.7	59.3	61.2	57.9	52.0
036	Insurance & Finance	18.9	58.6	62.6	52.4	48.9
037	Construction & Repair	10.4	67.8	67.5	49.0	47.1
038	Telecommunications	57.6	54.5	45.1	60.8	60.1
039	Transport & Storage	21.3	62.5	63.1	54.8	47.6
040	Material Treatment	9.6	68.0	63.9	49.4	47.3
041	Education & Entertainment	35.0	61.1	60.6	54.2	52.5
042	Scientific & Tech Services	39.3	57.8	59.4	58.0	51.4
043	Hotels & Restaurants	9.1	62.9	58.7	53.8	48.1
044	Medical & Beauty Services	20.6	62.9	62.1	53.7	51.3
045	Legal & Personal Services	41.5	60.3	65.0	60.1	48.7

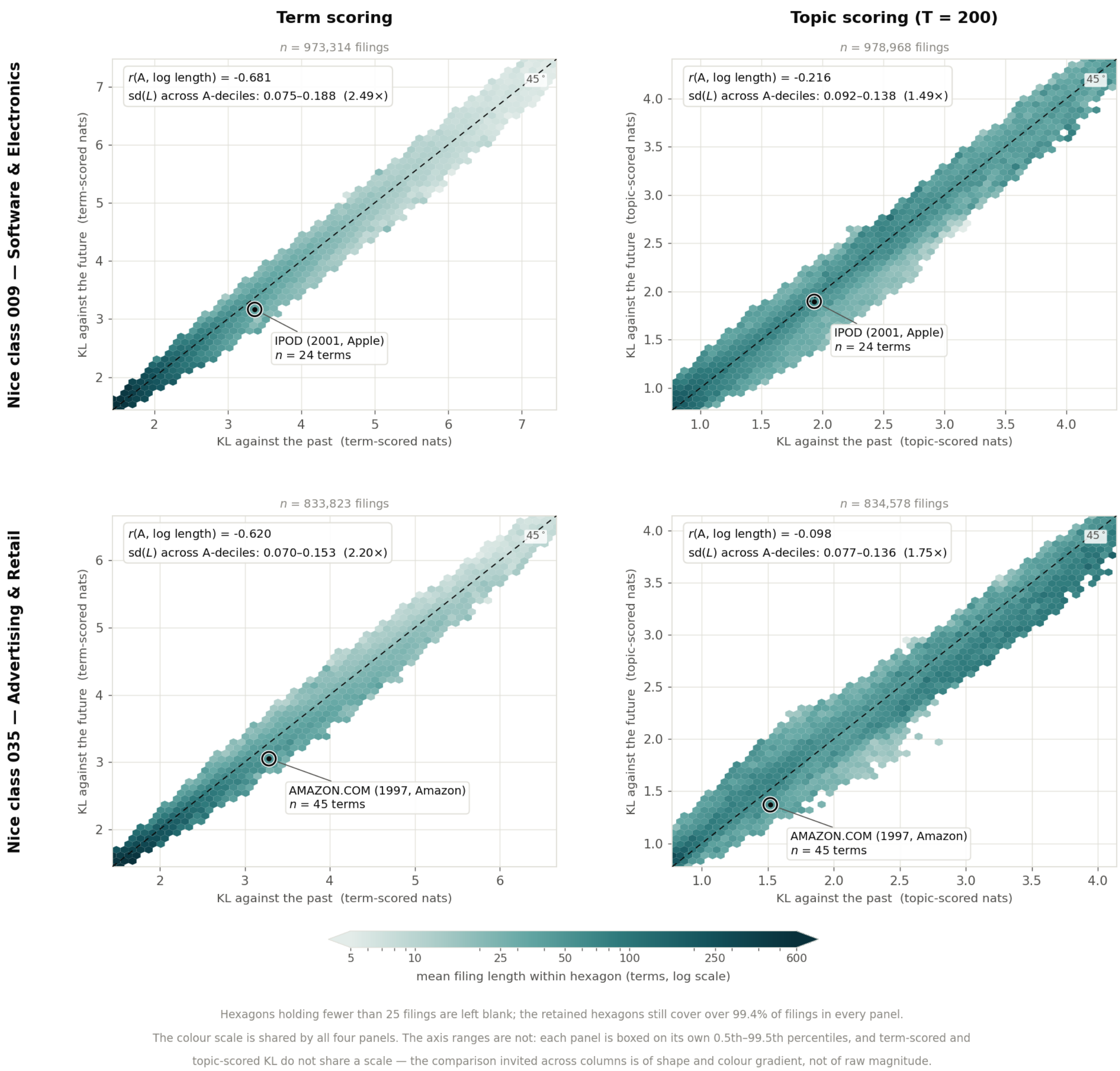

Figure S6: Mean filing length across the two axes, by representation and class, for filings 1995–2019 scored on per-filing reference windows. The horizontal axis is surprise against the preceding five years, the vertical axis surprise against the following five; a filing on the dashed diagonal is equally far from both, and distance below it is lead. Hexagons holding fewer than 25 filings are blank; the retained hexagons cover over 99.4% of filings in every panel. The colour scale is shared across panels; the axis ranges are not, since term-scored and theme-scored KL do not share a scale, axis ranges differ across columns because term- and theme-scored KL are not on a common scale, and only shape and colour gradient are comparable. Both columns are drawn on the filings the per-filing windows score, which differ between representations by under one percent.

Table S2: Top theme loading for three filings, by representation. Term scoring resolves the novel content of all three; theme scoring resolves none of it at any tested resolution.

	AMAZON.COM (1997)	KOMBUCHA (1997)	ETHEREUM (2018)
novel content (term)	line search, search ordering	kombucha, saccharose	blockchains, distributed computing
$T = 50$	video, computer, music (0.40)	beverages, fruit, drinks (0.64)	software, online, downloadable (0.33)
$T = 200$	entertainment, video, games (0.29)	beverages, drinks, coffee (0.60)	field, services, development (0.37)
$T = 500$	computer, software, data (0.22)	beverages, drinks, fruit (0.61)	software, online, computer (0.49)

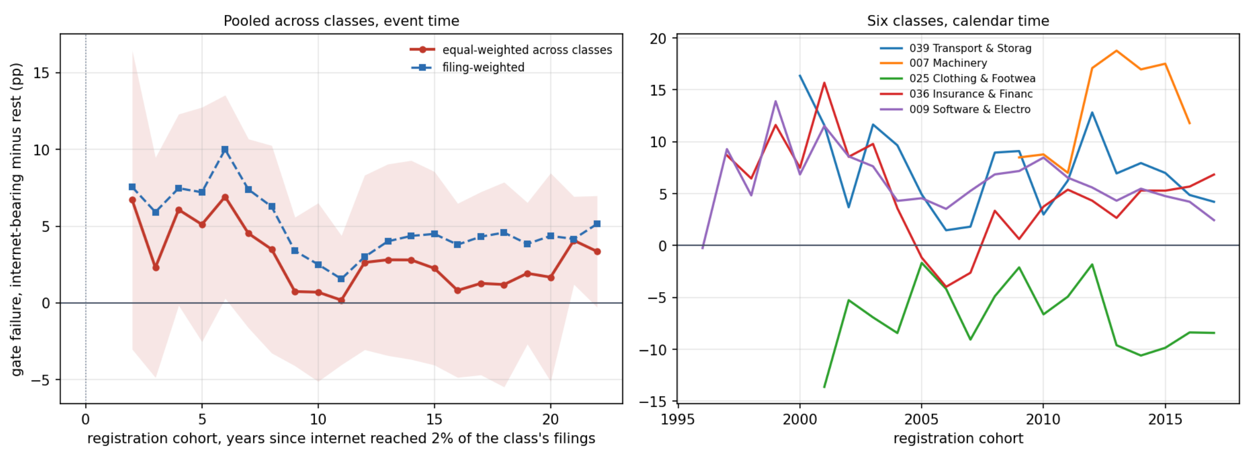

Figure S7: The internet gap in failure at the five-year proof, inside each class. *Left:* internet-bearing minus other registrations' failure rate, by registration cohort in years since the internet pattern reached 2% of the class's filings, averaged across classes (band: one standard deviation across classes). *Right:* five classes in calendar time. Cells with fewer than 100 registrations in either group are dropped.

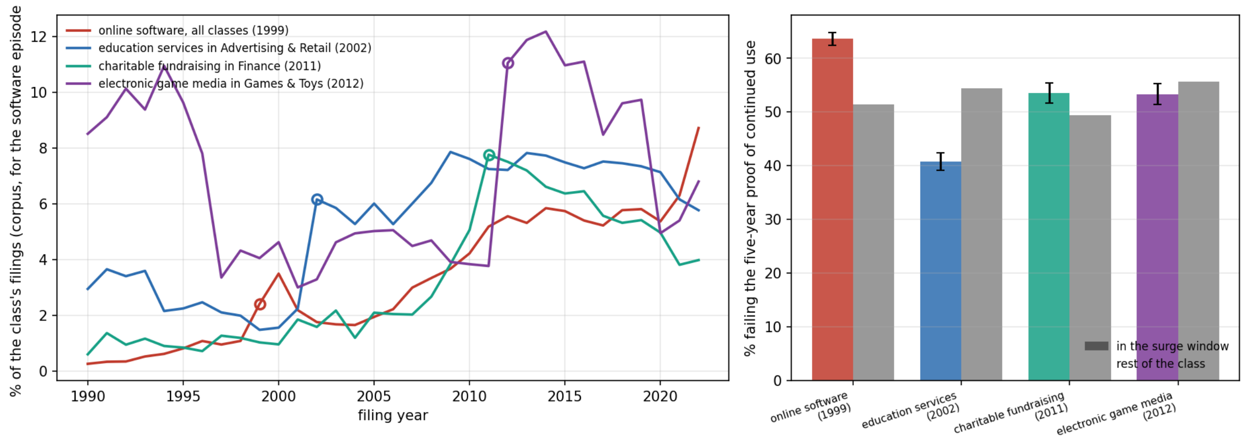

Figure S8: Four surge episodes under the 500-theme model. *Left:* the theme's share of its class's filings by filing year (for the online software episode, of all filings corpus-wide), on the 25% systematic sample the surge screen uses; the open circle marks the surge year. *Right:* failure at the five-year proof for registrations whose dominant theme was the surging one, filed in the three-year surge window (colored, 95% intervals), against the rest of the same class in the same cohorts (grey).

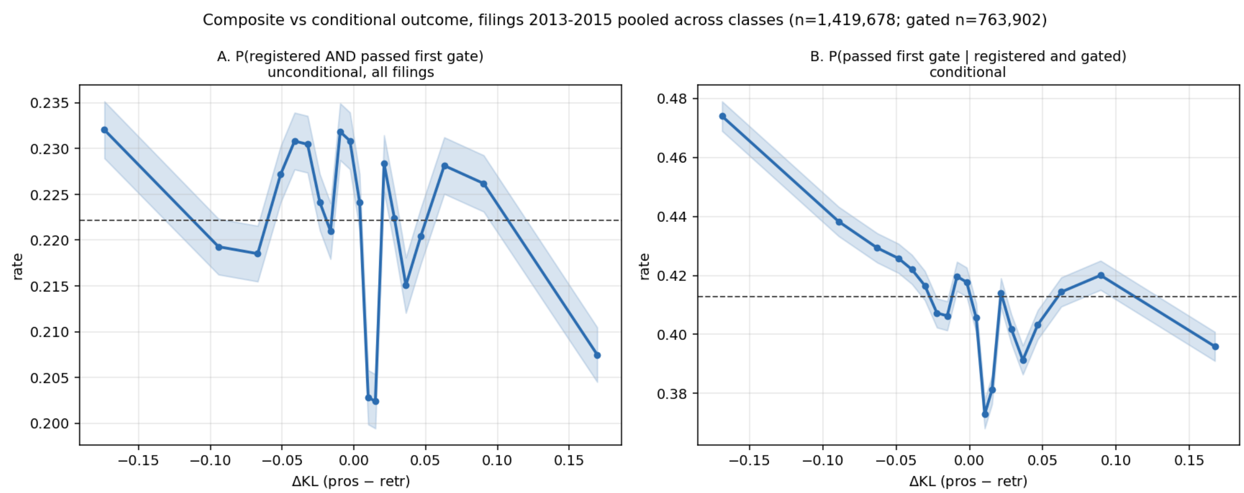

Figure S9: Filings 2013–2015 pooled across classes, theme-scored on per-filing reference windows ($n = 1{,}419{,}678$; gated subset $n = 763{,}902$). *A:* the unconditional composite P(registered and passed the five-year proof) over all filings. *B:* P(passed | registered and gated) on the same filings. The unconditional view is nearly flat in lead (22.4% to 22.1% across quintiles); conditioning on grant is what exposes the gradient.

Table S3: Pair-lead contrasts at the five-year proof by class: settled registrations, top-minus-bottom pair-lead quintile within class and registration year, raw and net of lead and atypicality.

Class	n	Raw Q5−Q1 (pp)	Net of L, A (pp)	SE
001	55,152	−0.74	+0.09	0.66
002	16,021	+1.49	+1.43	1.23
003	118,244	−0.58	−0.83	0.46
004	16,656	+0.23	−1.21	1.22
005	124,864	−1.63	+0.94	0.45
006	44,119	+0.70	+0.91	0.74
007	74,422	+1.07	+2.00	0.57
008	25,959	+0.45	+0.20	0.97
009	452,797	+1.74	+1.63	0.23
010	70,390	+1.61	+1.74	0.59
011	75,362	+0.87	+0.84	0.57
012	55,717	+0.43	−0.06	0.67
013	9,224	+0.09	+3.06	1.59
014	55,300	+2.50	+3.78	0.67
016	167,676	−3.83	−2.18	0.39
017	23,840	+0.01	+0.97	1.00
018	58,168	+1.32	+0.03	0.65
019	34,311	+1.19	+1.39	0.85
020	63,624	−0.41	+0.25	0.63
021	66,971	+0.17	+1.24	0.61
022	9,347	+4.16	+2.07	1.62
024	32,491	+2.26	+3.22	0.88
025	252,013	−0.86	+1.00	0.31
026	13,819	−4.70	−4.59	1.34
027	10,195	+0.06	+0.65	1.56
028	111,782	−0.83	−0.05	0.47
029	60,867	+3.63	+0.48	0.64
030	102,559	+0.65	+0.32	0.49
031	32,280	−2.49	−0.74	0.87
032	48,320	−0.81	−1.21	0.72
033	56,944	+3.00	+4.92	0.66
034	16,060	+1.57	+6.40	1.24
035	428,522	−1.62	−2.09	0.24
036	182,966	+0.70	+0.21	0.37
037	86,313	+1.32	+0.48	0.54
038	64,306	+0.13	−0.57	0.61
039	54,581	+0.04	−0.01	0.68
040	43,934	+1.39	+0.30	0.75
041	381,442	+1.19	+1.11	0.25
042	268,273	−2.41	+1.27	0.30
043	83,647	−0.90	−0.66	0.55
044	89,795	+0.42	+0.20	0.53
045	62,033	+3.38	+3.28	0.63

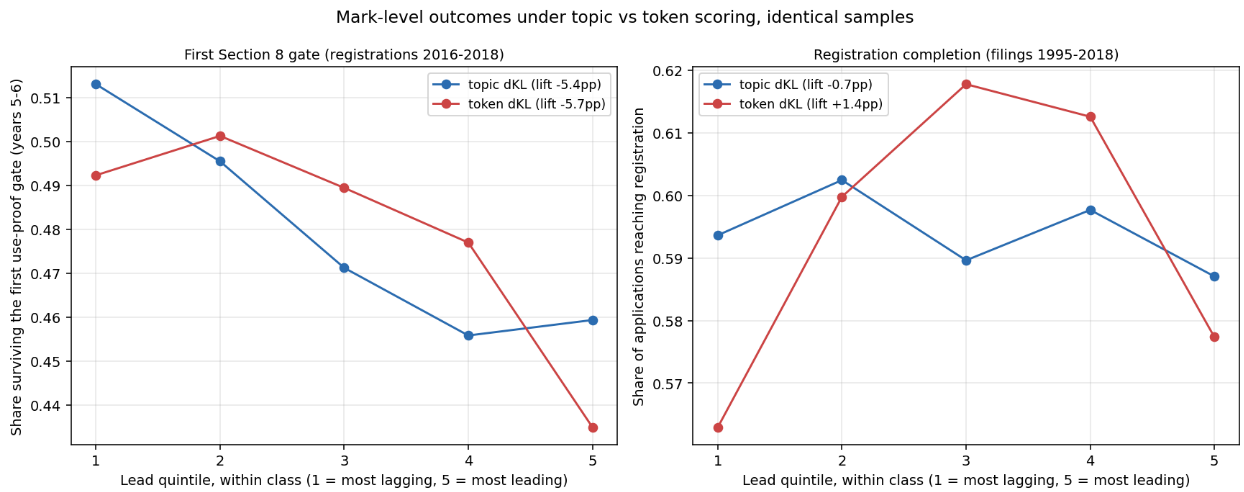

Figure S10: The two mark-level outcomes under theme and term scoring on identical samples, pooled across classes, by within-class lead quintile (1 most lagging, 5 most leading). *Left:* the share of registrations 2016–2018 that survived the five-year proof of continued use at years five to six, read from terminal status codes, where the two representations agree closely. *Right:* registration completion (filings 1995–2018), where they do not: the inverse-U in the signed axis is present under term scoring (depth +0.048) and absent under theme scoring (depth −0.001). The gate result is scoring-robust; the registration curvature on this axis is not, and Section 4 reports only the unsigned version.

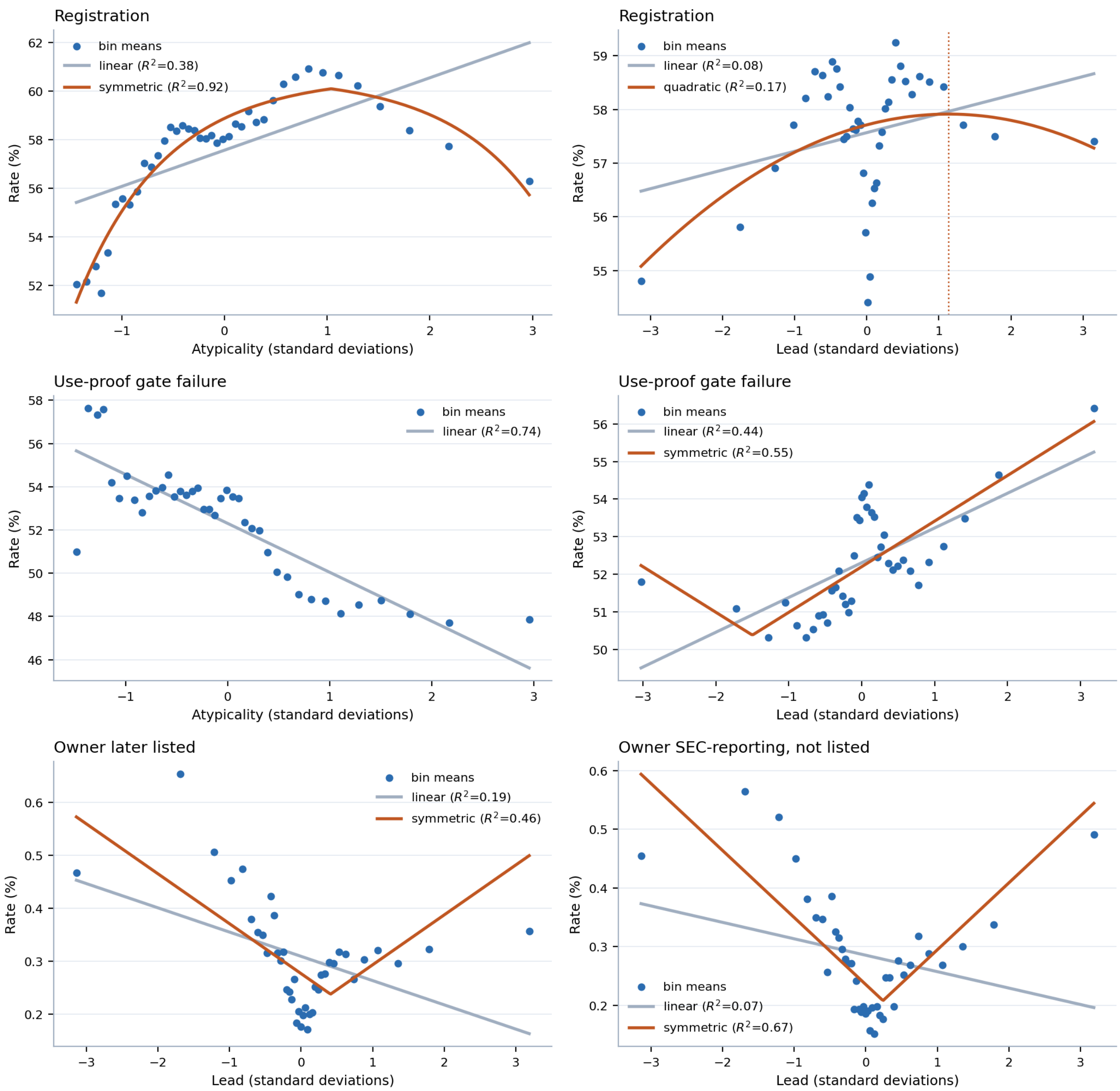

Figure S11: Binned profiles at the three gates, with the linear reading the paper's contrasts assume and the shape selected by AIC where that is not linear. Forty equal-count bins; the dotted line marks the fitted vertex. The bottom row is the case that matters: a linear fit through the public-reporting profile slopes downward while the data turn up at both tails.

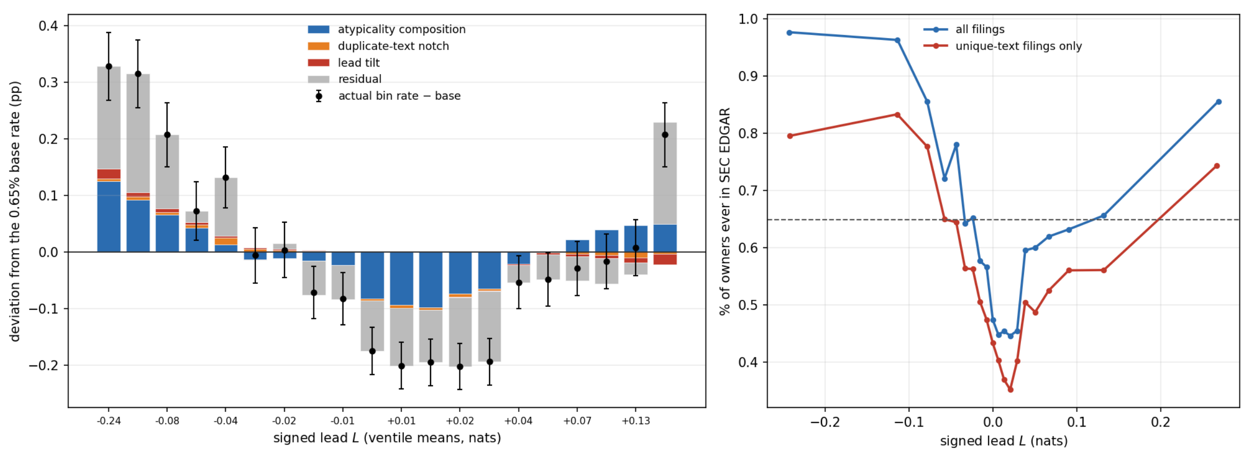

Figure S12: The signed-lead listing profile taken apart. *Left:* each ventile's deviation of P(owner ever in SEC EDGAR) from the 0.65% base, on 2.07 million registered debuts 1995–2018, split into the part predicted by atypicality composition alone (50 quantile bins of A), the part added by an exact-duplicate-text indicator, a linear tilt in L fitted to the remainder, and the residual; bars stack from zero, dots are the actual deviations with 95% intervals. *Right:* the same profile on all filings and on filings whose normalized description text is unique in their class; the central dip survives, and deepens, without duplicated text.